Lecture 12: The Integral Test

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Based on Thomas' Calculus, 15th Edition in SI Units, Pearson Education Limited

Sem II, 2025/2026

Learning Outcome

At the end of this lecture, you should be able to:

- ☐ Understand when the Integral Test can be used.
- ☐ Use the Integral Test to decide if a series converges or diverges.
- ☐ Use the error bounds to estimate the partial sums.

Nondecreasing Partial Sum

↑ sequence of partial sum $\{s_n\} = \{s_1, s_2, s_3, \ldots\}$

Suppose that $\sum_{n=1}^{\infty} a_n$ is an infinite series with $a_n \geq 0$ for all n. Then each partial sum is greater or equal to its predecessor.

$$s_n = a_1 + a_2 + \cdots + a_n$$

$$s_{n+1} = a_1 + a_2 + \cdots + a_n + a_{n+1}$$

$$= s_n + a_{n+1} \geq s_n \quad \text{since} \quad a_{n+1} > 0.$$

A series $\sum_{n=1}^{\infty} a_n$ of nonnegative terms converges if and only if its partial sums are bounded from above.

proof: ($\Rightarrow$) Suppose that $\sum a_n$ converges to L.

Since $a_n \geq 0$ for all n, $s_{n+1} \geq s_n$ for all n.

Therefore, $\{s_n\}$ must be non-decreasing and bounded from above by $\lim_{n\to\infty} s_n = L$.

($\Leftarrow$) Suppose that $\{s_n\}$ is bounded from above by L.

Again, since $a_n \geq 0$ for all n,

$s_{n+1} \geq s_n$ for all n.

$\{s_n\}$ is non-decreasing.

$\{s_n\}$ is bounded from above by L,

$\{s_n\}$ is bounded from below by a_1.

So $\{s_n\}$ is monotonic and bounded.

By Monotonic Sequence Theorem,

$\sum_{n=1}^{\infty} a_n$ is convergent. ($\{s_n\}$ converges)

Nondecreasing Partial Sum

Example: The harmonic series $\sum_{n=1}^{\infty} \frac{1}{n}$ diverges.

$$\sum \frac{1}{n} = 1 + \frac{1}{2} + \left(\frac{1}{3} + \frac{1}{4}\right) + \left(\frac{1}{5} + \frac{1}{6} + \frac{1}{7} + \frac{1}{8}\right) + \left(\frac{1}{9} + \dots + \frac{1}{16}\right) + \dots$$

$$\geq 1 + \frac{1}{2} + \left(\frac{1}{4} + \frac{1}{4}\right) + \left(\frac{1}{8} + \frac{1}{8} + \frac{1}{8} + \frac{1}{8}\right) + \left(\frac{1}{16} + \dots + \frac{1}{16}\right) + \dots$$

$$\geq 1 + \frac{1}{2} + \frac{1}{2} + \frac{1}{2} + \frac{1}{2} + \dots$$

Let define our partial sum as below.

the usual partial sum.

$S_{2^0} = S_1' = 1 \geq 1$ (1 + 0 times of $\frac{1}{2}$)

$S_{2^1} = S_2' = 1 + \frac{1}{2} \geq 1 + \frac{1}{2}$ (1 + 1 times of $\frac{1}{2}$)

$S_{2^2} = S_3' = 1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} \geq 1 + \frac{1}{2} + \frac{1}{2}$ (1 + 2 times of $\frac{1}{2}$)

$S_{2^3} = S_4' = 1 + \frac{1}{2} + \frac{1}{5} + \frac{1}{4} + \frac{1}{5} + \frac{1}{6} + \frac{1}{7} + \frac{1}{8} \geq 1 + \frac{1}{2} + \frac{1}{2} + \frac{1}{2}$

$S_{2^4} = S_5' = 1 + \frac{1}{2} + \dots + \frac{1}{16} \geq 1 + \frac{1}{2} + \frac{1}{2} + \frac{1}{2} + \frac{1}{2}$

$S_k' = 1 + \frac{1}{2} + \dots + \frac{1}{2^{k-1}}$

$$S_{2^j} = S_{2^{j+1}} \geq 1 + \underbrace{\frac{1}{2} + \cdots + \frac{1}{2}}_{j \text{ times.}}$$

If $n = 2^j$, $S_n > 1 + \frac{j}{2}$

As $j \to \infty$, $1 + \frac{j}{2} \to \infty$

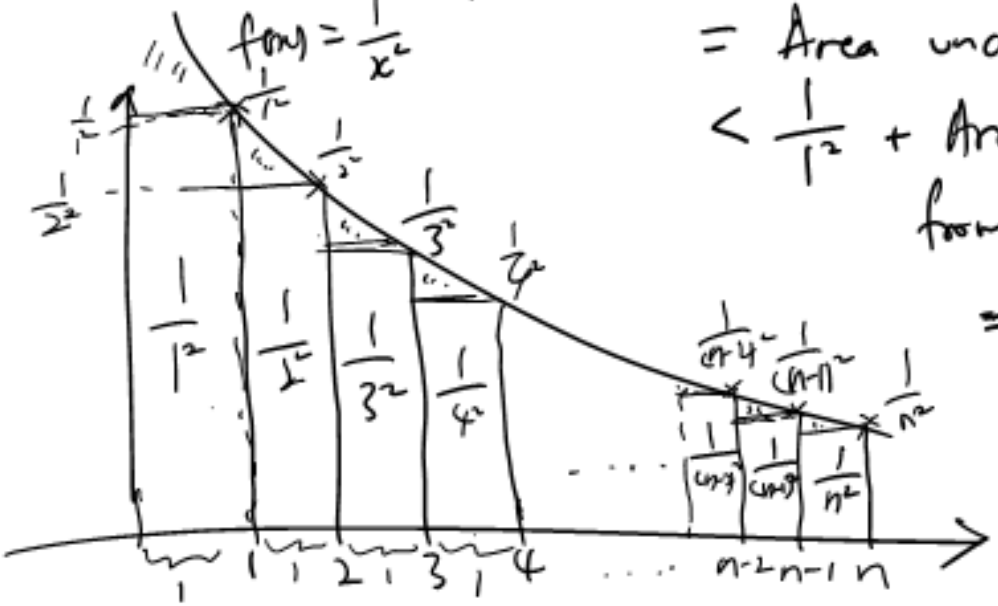
$\therefore$ the sequence of partial sum $\{S_n\}$ is not bounded from above, so the series must be divergent.

The Integral Test

Does the series $\sum_{n=1}^{\infty} \frac{1}{n^2}$ converge?

$$1 + \frac{1}{2^2} + \frac{1}{3^2} + \dots + \frac{1}{k^2} + \dots$$

Let $S_n = \sum_{k=1}^{n} \frac{1}{k^2} = \frac{1}{1^2} + \frac{1}{2^2} + \frac{1}{3^2} + \dots + \frac{1}{n^2}$



= Area under of rectangles

$< \frac{1}{1^2}$ + Area under the curve $f(x) = \frac{1}{x^2}$ from 1 to n.

$= 1 + \int_1^n \frac{1}{x^2}\,dx$

As $n \to \infty$, $S_n = 1 + \lim_{n\to\infty} \int_1^n \frac{1}{x^2}\,dx$

$= 1 + \lim_{n\to\infty} \left[\frac{-1}{x}\right]_1^n$

$$= 1 + \left[\underbrace{\lim_{n \to \infty} \frac{-1}{n}}_{\to 0} - (-1) \right]$$

$$= 2$$

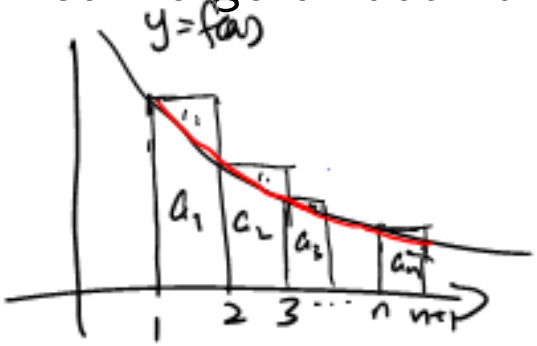
(the seq of partial sum is bounded from above by 2)

So $\sum \frac{1}{n^2}$ converges.

The Integral Test

* $\sum a_n$ is not necessarily equal to $\int_N^\infty f(x)\,dx$.

Let $\{a_n\}$ be a sequence of positive terms. Suppose that $a_n = f(n)$, where f is a (1) continuous, (2) positive, (3) decreasing function of x for all $x \geq N$ where N is a positive integer. Then the series $\sum_{n=N}^{\infty} a_n$ and the integral $\int_N^\infty f(x)\,dx$ both converge or both diverge.

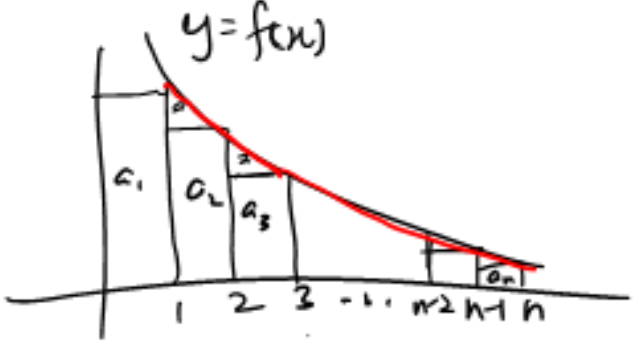


partial sum $S_n = a_1 + a_2 + \dots + a_n$

= the sum of the area of rectangles.

$> \int_1^{n+1} f(x)\,dx$

↖ the area under the curve



partial sum $S_n = a_1 + a_2 + \dots + a_n$

= the sum of area of all rectangles.

$< a_1 + \int_1^n f(x)\,dx$

So, $\int_1^{n+1} f(x)\,dx < S_n < a_1 + \int_1^n f(x)\,dx$

Now, if $\int_1^\infty f(x)\,dx$ converges. $\Rightarrow a_1 + \int_1^n f(x)\,dx$ converges as $n \to \infty$

$\Rightarrow \{S_n\}$ is bounded from above $\Rightarrow \sum a_n$ converges.

On the other hands,

if $\int_1^\infty f(x)\,dx$ diverges $\Rightarrow$ $\int_1^{n+1} f(x)\,dx$ diverges

$c_3 \; n \to \infty$

$\Rightarrow \{s_n\}$ is unbounded and $\sum a_n$ diverges.

The Integral Test

Example: For a real constant p, the p-series

$$\sum_{n=1}^{\infty} \frac{1}{n^p} = \frac{1}{1^p} + \frac{1}{2^p} + \frac{1}{3^p} + \cdots + \frac{1}{n^p} + \cdots$$

converges if $p > 1$ and diverges if $p \leq 1$.

$p=1$: $\sum \frac{1}{n}$ is a divergent harmonic series.

$p>1$: $\int_1^{\infty} \frac{1}{x^p}\,dx = \lim_{b\to\infty} \left[\frac{x^{-p+1}}{-p+1}\right]_1^b$

x^{-p}

$= \lim_{b\to\infty} \frac{1}{1-p} \cdot \left(\frac{1}{b^{p-1}}\right) - \left(\frac{1}{1-p} \cdot 1\right)$ ($\to 0$)

$= -\frac{1}{1-p}$ (converges)

Since $f(x) = \frac{1}{x^p}$ is decreasing, cts and positive function from 1 to ∞ for $p>1$, by the Integral Test, the series converges.

$p \leq 1$: When $p \leq 0$ $\lim_{n\to\infty} \frac{1}{n^p} = \lim_{n\to\infty} n^{-p}$

$= \lim_{n\to\infty} n^{q}$, $q \geq 0$

$\to \infty$ as $n \to \infty$.

So the series diverges by the nth term test for divergence.

When $0 < p < 1$, $\int_0^1 \frac{1}{x^p}\,dx = \lim_{a\to 0^+} \int_a^1 \frac{1}{x^p}\,dx$

$= \lim_{a\to 0^+} \left[\frac{1}{1-p} \cdot \left(\frac{1}{x^{p-1}}\right)\right]_a^1$

$= \frac{1}{1-p} - \lim_{a\to 0^+} \frac{1}{1-p}\left(\frac{1}{a^{p-1}}\right)$ $\to -\infty$

(diverges)

So, $\sum \frac{1}{n^p}$ diverges by the Integral Test.

So, $\sum \frac{1}{n^p}$ converges when $p > 1$ and diverges when $p \leq 1$.

The Integral Test

Example: The series $\sum_{n=1}^{\infty} \frac{1}{n^2+1}$ converges.

Let $f(x) = \frac{1}{x^2+1}$, then f is cts, positive and non-decreasing on $[1, \infty)$

$$\int_1^{\infty} f(x)\,dx = \lim_{b\to\infty} \int_1^b \frac{1}{x^2+1}\,dx$$

So by the Integral Test, $\sum_{n=1}^{\infty} \frac{1}{n^2+1}$ converges.

$$= \lim_{b\to\infty} [\arctan x]_1^b$$

Again, this does not mean that $\sum \frac{1}{n^2+1} \to \pi/4$

$$= \lim_{b\to\infty} \arctan b - \arctan 1$$

$$= \pi/2 - \pi/4 = \pi/4 \quad \text{(converges)}$$

The Integral Test

Example: The series $\sum_{n=2}^{\infty} \frac{1}{n(\ln n)^2}$ converges.

(Exercise)

$$\int_2^{\infty} \frac{1}{x(\ln x)^2}\,dx = \ln 2$$

Error Estimation

Suppose a series $\sum a_n$ with positive terms converges to S, we want to estimate the size of the remainder R_n by measuring the difference between S and the nth partial sum s_n.

$\sum_{n=1}^{\infty} a_n = \lim_{n\to\infty} s_n$

$$R_n = S - s_n = a_{n+1} + a_{n+2} + a_{n+3} + \cdots$$

Let f be a function such that $f(n) = a_n$

Similar to the proof of the integral test,

$$R_n = a_{n+1} + a_{n+2} + a_{n+3} + \cdots \leq \int_n^{\infty} f(x)\,dx$$

and $$R_n = a_{n+1} + a_{n+2} + a_{n+3} + \cdots \geq \int_{n+1}^{\infty} f(x)\,dx$$

Bounds for the Remainder in the Integral Test

Suppose $\{a_k\}$ is a sequence of positive terms with $a_k = f(k)$, where f is a continuous, positive, decreasing function of x for all $x \geq n$, and suppose that $\sum a_n$ converges to S. Then the remainder $R_n = S - s_n$ satisfies the inequalities

$$\int_{n+1}^{\infty} f(x)\,dx \leq R_n \leq \int_{n}^{\infty} f(x)\,dx.$$

$R_n = S - s_n$

To estimate S, we add the partial sum s_n to each side of the inequalities.

$$s_n + \int_{n+1}^{\infty} f(x)\,dx \leq S \leq s_n + \int_{n}^{\infty} f(x)\,dx.$$

Bounds for the Remainder in the Integral Test

Example: Estimate the sum of the series $\sum \frac{1}{n^2}$ using the previous inequalities and $n = 10$. (We are asked to estimate $\sum \frac{1}{n^2}$ by using S_{10})

$$a_n = \frac{1}{n^2}, \quad [n = 10]$$

$$S_{10} + \int_{11}^{\infty} \frac{1}{x^2}\,dx \leq S \leq S_{10} + \int_{10}^{\infty} \frac{1}{x^2}\,dx$$

$$S_{10} = \frac{1}{1^2} + \frac{1}{2^2} + \cdots + \frac{1}{10^2} \approx 1.54977$$

$$\int_{11}^{\infty} \frac{1}{x^2}\,dx = \lim_{b\to\infty}\left[-\frac{1}{x}\right]_{11}^{b} = \lim_{b\to\infty}\left(-\frac{1}{b}\right) - \left(-\frac{1}{11}\right)$$

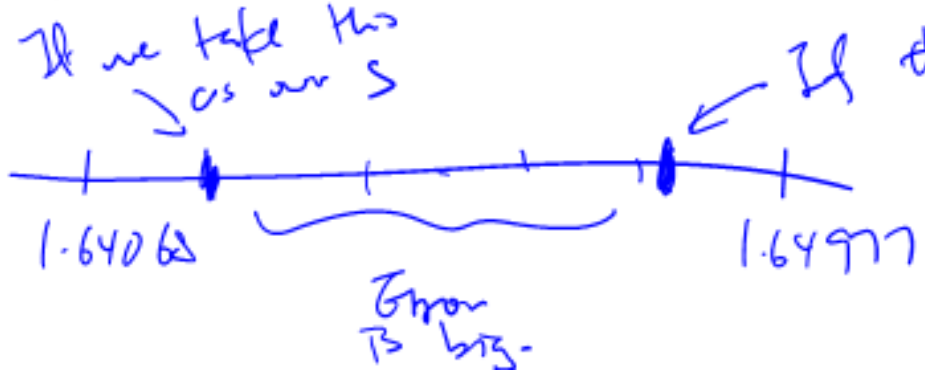
$$= \frac{1}{11}$$

$$\int_{10}^{\infty} \frac{1}{x^2}\,dx = \frac{1}{10}$$

$$1.64068 \approx 1.54977 + \frac{1}{11} \leq S \leq 1.54977 + \frac{1}{10} \approx 1.64977$$

To estimate the S with the minimal error bound, we take the midpoint.

$$\sum_{n=1}^{\infty} \frac{1}{n^2} \approx 1.6452$$



So to minimise the error, we take the midpoint